

eSpark Consulting Group

Project Proposal

Parking Barrier

Project Overview

This project aims to develop an automated barrier system that integrates remote control and monitoring capabilities via Bluetooth Low Energy (BLE) and TCP/IP communication. The system consists of a Barrier module, which controls and monitors individual barriers, and a Gateway module, which aggregates data from multiple barriers and communicates with a central server. The system is designed for high durability in industrial environments, with an expected battery life of 6-8 months for the Barrier module.

Key Requirements and Features

Barrier Control Capabilities:

- **Remote and Multi-Access Control:** Supports remote operation via BLE for single or multiple barriers simultaneously, with user permissions allowing only authorized personnel access. This feature is especially useful for facilities needing synchronized control across multiple entry points.

- **Smart Vehicle Detection and Response:** Features configurable vehicle detection with advanced settings, including timed automation and adaptive responses to traffic density. The system can learn from peak usage hours to optimize barrier response, reducing wait times.

- **Real-Time Health Diagnostics and Analytics:** Offers comprehensive health diagnostics that track metrics beyond hardware status—such as temperature, signal quality, and operational cycles—to provide analytics that optimize system performance and forecast maintenance needs**.**

- **Battery Monitoring with Eco-Mode:** Includes an eco-mode that intelligently reduces battery usage during low-traffic periods, extending battery life and lowering operational costs.

Gateway Specifications:

- **Efficient Data Aggregation and Relay with Batch Reporting:** Aggregates data from up to 20 barriers, transmitting consolidated reports to the central server during off-peak hours or intervals as configured. This reduces network load while ensuring critical, real-time data is still available.
- **Advanced Data Processing with BLE to TCP/IP Conversion:** Translates BLE data from barriers into TCP/IP packets with lossless accuracy, supporting both real-time data flow and batch processing. The gateway's data caching ensures data integrity during transmission, even under high network loads.
- **Resilient Industrial-Grade Build:** Engineered to handle vibrations, dust, and moisture in extreme environments, the gateway's enclosure is IP-rated for enhanced durability, making it ideal for outdoor and heavy-duty industrial applications
- **Redundant Power Management:** Built with an uninterruptible power source and a secondary power backup, ensuring operational continuity even during primary power source outages. The system sends alerts when backup power is activated, enabling timely power management
- **Scalable Centralized Control:** Integrates with cloud-based or on-premises control systems, enabling management of hundreds of barriers across multiple locations through one interface. This feature is ideal for large facilities or companies operating across multiple sites.

Enhanced Security Features:

- **Tamper Detection and Alert System:** Incorporates sensors to detect physical tampering with the barrier or gateway, automatically sending an alert to the central server and locking control functions to prevent unauthorized access.

Hardware and Durability Specifications:

- **Compact, Modular Barrier Board Design:** Compact 8 cm x 8 cm board design that easily fits within barrier housing, with modular components for easy upgrades or replacements. The board uses industrial-grade, high-resilience materials to protect against moisture, dust, and high-impact scenarios.
- **Enhanced Battery Performance with Energy Management Algorithms:** Powered by a high-capacity battery designed to last 8-12 months, with energy management algorithms that monitor usage and optimize power for high-demand and low-demand periods, extending battery life.
- **Quick-Swap Maintenance Components:** Designed with modular parts and quick-release connectors for rapid on-site repairs or component swaps. Maintenance staff can quickly access diagnostic ports to check component health without dismantling the system.

- **User-Friendly LED Indicators and Status Display:** Equipped with status LEDs and a small digital display on the gateway unit to show real-time operational status, power level, and connectivity, enabling users to quickly assess the system's health.

Milestone Breakdown

Milestone 1: Research Analysis and Parts selection (Barrier Electronics and Gateway Electronics)

Timeline: **5 Working days**
Budget: **$500**
Deliverables:
- Full system Block Diagram including major selection part numbers
- Excel sheet with reference links to selected part number

Milestone 2: Schematic Design Barrier Electronics

Timeline: **5 Working days**
Budget: **$400**

Milestone 3: Schematic Design Gateway Electronics (This milestone can run in parallel with M2)

Timeline: **4 Working days**
Budget: **$350**

Milestone 4: PCB Design Barrier Electronics

Timeline: **10 Working Days**
Budget: **$650**

Milestone 5: PCB Design Gateway Electronics

Timeline: **8 Working Days**
Budget: **$550**

Milestone 6: Prototyping and Testing

Timeline: **4 Working Days**
Budget: **$150**

Milestone 7: Main Application Firmware Development Barrier Electronics

Timeline: **4 to 5 Working Days**
Budget: **$500**

Milestone 8: Main Application Firmware Development Gateway Electronics

Timeline: **8 Working Days**
Budget: **$800**

Conclusion

- This proposal outlines an efficient barrier control solution with an integrated gateway system for reliable remote management and diagnostics. This phased approach, incorporating rigorous testing and quality assurance, ensures robust performance, enabling the solution to handle industrial environments effectively.